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RESEARCH ARTICLE

Convergence Innovation: Empowering the New Quality Development of Higher Education with Digital and Intelligent Technologies*

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Abstract

Higher education, serving as the primary source of talent, the principal platform for research, and the chief engine of innovation, is a key force in accelerating the development of new quality productive forces. As a new strategy for empowering the new quality development of higher education through digital and intelligent technologies, convergence innovation combines theoretical thinking, research paradigms, and practical innovations. This study argues that the convergence innovation research and convergence innovation education should be recognized as a new frontier of interdisciplinary research and a new paradigm for educational innovation enabled by digital and intelligent technologies. In addition, this study explores the connotative elements, logical mechanisms, and implementation pathways of convergence innovation research and convergence innovation education. Furthermore, it proposes strategies and recommendations to advance convergence innovation in Chinese higher education. These include cultivating convergence innovation talent through convergence innovation research and convergence innovation education, fostering a science of team science (SciTS)-based convergence innovation culture, and improving the governance framework for convergence innovation in higher education.

Keywords new quality productive forces; convergence innovation; convergence innovation education; design research; convergence innovation culture

1 Introduction

The State Council of the People's Republic of China issued the *New Generation Artificial Intelligence Development Plan* in 2017, which advocated for launching artificial intelligence (AI) interdisciplinary exploratory research and promoting the intersection and convergence of AI with related basic disciplines. Since then, China has emphasized the integrated development of AI and education. A series of policy documents, such as the *AI Innovation Action Plan for Institutions of Higher Education*, *Guidance on Accelerating the Construction*

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of "Double First-Class" Universities, and *Several Opinions on the Construction of "Double First-Class" Universities to Promote the Integration of Disciplines and Accelerate the Cultivation of Postgraduates in the Field of Artificial Intelligence* have been promulgated, highlighting interdisciplinary integration and the targeted cultivation of advanced AI talent. Under the guidance of policy, new interdisciplinary fields have been established to focus on cutting-edge science and technology areas, including new energy, integrated circuit science and engineering, AI, and others. Higher education institutions (HEIs) have reached a strategic consensus on building cultivation systems that equally emphasize foundational theory and "AI + X" interdisciplinary talent development. Moreover, they are committed to deeply exploring integrated mechanisms for discipline construction and talent cultivation, innovating disciplinary organization models and curricula, and expanding interdisciplinary educational functions.

The overall quantity of interdisciplinary research in China is considerable. However, the high-quality and far-reaching research results are relatively lacking. This stems from the long-standing system of disciplinary compartmentalization, which has deepened the divide between the natural sciences, social sciences, and the humanities. This study introduces emerging paradigms such as convergence innovation education and research and advocates for the integration of individual knowledge, cognition, and thinking. It aims to dismantle the boundaries separating disciplines, research, and application, and foster collaboration among individuals, machines, and the environment. Moreover, it explores novel trajectories in convergence innovation theory, research, and governance, thereby securing the strategic and sustainable development of convergence innovation projects. This approach is to address scientific challenges and complex societal issues while highlighting the pivotal role of higher education in accelerating the development of new quality productive forces.

2 Convergence Innovation Research: A New Paradigm in Interdisciplinary Research

2.1 Origin and Significance

The concept of convergence innovation research traces its origins to technology convergence projects. At the beginning of the 21st century, the U.S. Department of Commerce and the U.S. National Science Foundation jointly held a workshop on Converging Technologies for Improving Human Performance. It introduced the Nano-Bio-Info-Cogno (NBIC) converging technologies concept as a blueprint for transcending traditional disciplinary boundaries, unifying sciences, and integrating technologies for the first time (Roco & Bainbridge, 2003). NBIC converging technologies denote the synergy, complementarity, and integration of: (a) nanoscience and nanotechnology; (b) biotechnology and genetic engineering; (c) information technology, including AI and communications; and (d) cognitive science, including cognitive neuroscience, aimed at achieving an improvement in human abilities, societal outcomes, the nation's productivity, and the quality of life (Wang & Ma, 2016). The report *Converging Technologies for Improving Human Performance* articulated the synergistic interplay across four key domains: "If the cognitive scientists can think it, the Nano people can build it, the Bio people can implement it, and the IT people can monitor and control it" to jointly promote science and technology innovation. This articulation underscores the strength of interdisciplinary cooperation while exemplifying how interdisciplinary complementarity and integration facilitate the improvement of technology and human abilities (Lü & Tan, 2023). Interdisciplinary convergence has opened new avenues for addressing complex problems and accelerated science and technology progress and knowledge innovation. It highlights the importance of methodological innovation in interdisciplinary cooperation and innovation while demonstrating the central role of integrative thinking in convergence innovation research for advancing social development.

If the 20th century is referred to as "the era of analysis," the 21st century has opened a new chapter as "the era of convergence." This shift signifies a transformation in research paradigms from discipline-specific, inward-looking analysis to interdisciplinary, integrative inquiry. The U.S. National Academy of Sciences states that convergence transcends disciplinary boundaries and integrates knowledge and technologies from Life Sciences, Physics, Chemistry, Mathematics, Computing, Engineering, and other relevant fields to form a comprehensive synthetic framework for tackling scientific and societal challenges. As an advanced form of cross-disciplines, convergence innovation hinges on the deep integration of ideas, methods, and technologies from diverse fields, generating novel solutions to complex problems and serving as a key strategy for addressing the challenges of emerging disciplines (United States National Research Council, 2015). The U.S. National Science Foundation has long prioritized interdisciplinary research, strategically promoting it by defining interdisciplinary focus areas, fostering interdepartmental collaboration, creating research centers, and supporting education. Through initiatives such as the "10 Big Ideas," "Growing Convergence Research," and "Convergence Accelerator," the U.S. National Science Foundation (2020) has steadily increased its support for convergence innovation research. Solving major challenges, such as those in human health, energy and resource scarcity, and space exploration, requires multidisciplinary convergence in ideas, methods, and technologies to spur innovation and creativity. Recognizing that convergence research has clear aims of innovation and creativity, this study proposes the concept of "convergence innovation research."

2.2 Theoretical Innovation

Convergence innovation research has drawn on an ensemble of theoretical traditions that collectively illuminate the dynamics of cross-domain knowledge integration. Among these, several foundational propositions carry particular explanatory relevance for higher education. The premise of structural isomorphism across knowledge, technology, and social systems (Bainbridge & Roco, 2016b) suggests that disciplinary boundaries, while operationally useful, do not reflect ontological divisions in the phenomena under study. For higher education, this implies that institutional structures organized around discrete disciplines may impede rather than facilitate intellectual progress on complex problems. Therefore, HEIs confront a structural tension between administrative convenience and epistemic integration.

This tension becomes more pronounced when examined through the lens of complex adaptive systems theory. HEIs, as paradigmatic complex systems, generate educational outcomes through dynamic interplays among curricula, pedagogies, technologies, and human agents. The emergent properties arising from these interactions, namely, institutional culture, innovative capacity, and graduate competencies, cannot be simplified to or predicted from any single component. Consequently, linear models of educational input and output consequently fail to capture the generative dynamics that distinguish transformative institutions from merely functional ones. Compounding this complexity, innovation trajectories follow path-dependent yet punctuated patterns, in which periods of incremental refinement alternate with episodes of discontinuous transformation. The integration of digital and intelligent technologies into higher education represents such a punctuation point, necessitating not merely the adoption of new tools but the reconceptualization of pedagogical relationships and institutional arrangements. These theoretical currents converge on a central insight, that is, addressing the multidimensional challenges confronting contemporary higher education requires transcending the analytical fragmentation that characterized 20th-century research paradigms.

2.3 Convergent–Divergent Thinking

Scientific inquiry and technological development co-evolve through alternating phases of convergence and divergence (Roco, 2016). Convergence is defined as the synthesis of previously disparate knowledge domains into coherent frameworks, whereas divergence refers to the subsequent differentiation of integrated knowledge into novel applications and specialized fields. This cyclical pattern provides an analytical lens for understanding innovation processes in higher education.

The convergence phase entails the confluence of ideas from multiple disciplines and their systematic integration into unified conceptual or methodological frameworks. In higher education contexts, this phase manifests in interdisciplinary curriculum design, cross-departmental research collaborations, and the synthesis of humanistic and technological approaches to teaching and learning. By contrast, the divergence phase involves transforming integrated knowledge into practical innovations and deriving new domains from these innovations. HEIs operationalize this through knowledge transfer, enterprise incubation, and the cultivation of graduates equipped with integrated competencies into diverse societal sectors.

This dynamic carries strategic implications for institutional governance. Effective innovation management requires mechanisms that facilitate integrative synthesis and generative differentiation. Overly rigid structures inhibit the fluid recombination that convergence demands, while excessively diffuse arrangements impede the focused application that divergence requires. The challenge lies in cultivating organizational forms sufficiently flexible to support iterative cycling between these complementary modes. Overall, the capacity to navigate such cycles constitutes a critical dimension of institutional resilience, namely, the ability not merely to absorb external perturbations but rather to transform disruptions into opportunities for systemic renewal.

2.4 Design Research to Promote Convergent Innovation Practice

Design epistemology posits that although design seemingly aims at creating objects, it in fact coordinates relationships among people, between people and objects, and between people and their environments, embodying a balance of pragmatism and idealism. As a form of "productive relations," design plays a major role in "guiding and adjusting" human–nature and human–society relationships and serves as a "convergent and integrative" innovation that drives transformations in socio-economic, science and technology, cultural, and educational domains. In educational research, the relationships among individuals, between individuals and their environments, and between schools and society are intertwined, giving rise to numerous tensions and dilemmas. Educators are encouraged to draw on design principles to coordinate relationships between different subjects and develop optimal solutions (Zhu & Dai, 2023). The mechanism by which design research promotes convergence innovation practice is summarized in four points:

First, design thinking is a form of "third-culture" wisdom, incorporating elements of complex systems engineering thinking. It is an integrative problem-solving approach used to situate problems, generate insights, and devise optimal solutions, Moreover, it is widely applied in contemporary design, engineering, and management. By embedding systems engineering-level understanding of interactivity, interdependency, and systematicness into the design process, it ensures solutions that are innovative, practical, and resilient within broader ecological contexts. Through its distinctive cognitive cycles of analysis and synthesis, divergence and convergence, design thinking demonstrates remarkable proficiency in identifying equilibria in complex systems. As a result, it endows outcomes with the robustness and adaptability sought in systems engineering.

Second, the creative emergence stage is a central domain of design thinking. Its key objective is to integrate novel values and practical logic and to deploy precise, effective methods that advance convergence innovation in practice. Design thinking, as a human-centered, innovative approach and mindset for tackling complex problems, aligns technological feasibility, business strategy, and user requirements to generate meaningful ideas that translate into social value. In educational research, design thinking fosters the emergence of creativity and stimulates researchers' innovative thinking by combining human-centered problem framing, iterative solution development, broad divergent ideation, and focused convergent refinement.

Third, design science research, as a novel problem-solving paradigm, is well-suited to addressing wicked problems. Since these problems are inherently complex, dynamic, and uncertain, they resist traditional linear solutions. Grounded in a distinct methodology, epistemology, and axiology, design science research employs strategies, including the solution-focused approach, iterative processes and feedback mechanisms, interdisciplinary cooperation, system thinking, and multi-party engagement to address wicked problems effectively. It demonstrates unique advantages and potential in confronting contemporary complex societal challenges.

Fourth, design research seeks to generate new knowledge by creating a dynamic, practice-based environment for discovering and resolving problems. Its exploratory character and iterative mechanism promote interdisciplinary knowledge integration and innovation and deepen a dynamic understanding of research topics. Consequently, it continuously challenges and expands existing epistemic boundaries in practice. The cyclical iteration of design research enables researchers to elevate concrete practical insights into generalizable theoretical knowledge, providing new perspectives and analytical frameworks for addressing complex problems.

Consequently, in the realm of educational research, design research, which is characterized by a human-centered concern, iterative development, interdisciplinary synthesis, and an innovation-driving orientation, tackles deeply rooted wicked problems. Moreover, through continuous knowledge generation and practice-based validation, it expands the frontiers of scholarship and social application, offering novel theoretical insights and practical pathways for addressing complex challenges.

2.5 AI-Empowered Shift in Convergence Innovation Research

A research paradigm is the basic system of beliefs formed in a given historical period that guides scientific research methods and theoretical interpretations. The scientific research paradigm has evolved from the traditional positivist paradigm established by Comte and Durkheim (practice-based research grounded in observation and theoretical deduction) to a theoretical paradigm (driven by hypotheses and deductive reasoning), then to a computational paradigm (based on computer simulation), and subsequently to a data paradigm (of data-intensive science), and it is now progressively shifting toward a fifth, intelligence-driven paradigm. Li Guojie, academician of the Chinese Academy of Engineering, termed “AI for Research (AI4R)” the fifth research paradigm, arguing that it inherits traditional scientific methods and promotes the exploration of intelligent systems, which emphasizes an organic integration of human cognition and AI (Cheng et al., 2020). AI powered by machine learning and deep learning exhibits exceptional capabilities in independent learning, deduction, and prediction, consistent with the logical mechanism and innovative attributes of convergence innovation research. AI4R is the AI-enabled paradigm shift in convergence innovation research, with six characteristics: (1) deep integration of AI across scientific research, technology development, and engineering practice, enabling automated knowledge generation and end-to-end intelligent research workflows; (2) deep human–machine integration, with emergent machine intelligence becoming an indispensable component in research, enables dark knowledge and generates machine-formulated conjectures; (3) a focus on complex systems, allowing effective response to problems of very high computational complexity; (4) an elevated role for probability theory and statistical inference in exploring non-deterministic scientific issues; (5) emphasis on interdisciplinary collaboration that promotes the organic integration of the four preceding research paradigms and the deep fusion of first principles thinking models with data-driven approaches; and (6) tighter coupling of scientific research and engineering practice as research increasingly depends on large-scale models and platforms (Li, 2024).

A review has comprehensively examined the development and application of AI in foundational scientific domains, including information science, mathematics, medicine, materials science, earth science, life sciences, physics, and chemistry. It has been concluded that AI-enabled convergence innovation research potentially delivers convergence innovation value to both scientific research and applications (Xu et al., 2021). For instance, in materials science, AI analyzes high-throughput material data to accelerate the innovation and design of new materials. In physics, AI algorithms efficiently simulate and predict particle dynamics in condensed matter. In chemistry, AI enables precise spectral analysis and the prediction of synthetic routes of compounds, speeding the development of new materials and medicines. In life sciences and medicine, AI's big-data analytics elucidates disease etiologies and supports personalized treatment planning. In mathematics, AI assists with question optimization and parameter estimation, enhancing model generalization and interpretability. Scientific breakthroughs, such as three-dimensional protein structure prediction, molecular dynamics simulation, and fully automated chip design, further illustrate that AI-enabled research improves efficiency while ensuring accuracy and has consequently become a major driver of scientific discovery and innovation.

3 Convergence Innovation Education: A New Paradigm Enabled by Digital and Intelligent Technologies

3.1 Convergence Innovation Education: An Educational Paradigm Shift Surpassing STEM

STEM (Science, Technology, Engineering, and Mathematics) education has emerged as a meta-discipline that dissolves traditional boundaries among science, technology, engineering, and mathematics. This interdisciplinary integration consolidates course content while prioritizing innovative thinking and the capabilities to solve complex, context-dependent problems. Through STEM education, students acquire the knowledge and skills required in the 21st century, which extend beyond disciplinary expertise to encompass creativity, problem solving, and critical thinking. The concept "new quality productive force" deeply reflects society's overall demand for innovation and creative capabilities. Therefore, STEM education transcends mere academic innovation to serve as a forward-looking strategy for addressing contemporary and future societal challenges.

Convergence innovation education has driven the cross-disciplinary integration of STEM, combining science, technology, engineering, and mathematics into a more application-oriented trans-discipline. As educational models evolve, STEM has expanded into a range of "STEM + X" fields, including STEAM (STEM + Arts), STEAMD (STEM + Arts, Drama), STREAM (STEM + Arts, Reading & Writing), STEAMD (STEM + Arts Design), STEMLE (STEM + Laws, Economics), and STREM (Science, Technology, Robotics, Engineering, and Multi-Media). In practice, the 5E Model of Instruction is regarded as an effective strategy for optimizing STEM education and providing personalized learning pathways. The 5E instructional model refined by Kennedy and Odell (2014) is based on the five phases of "engage, explore, explain, elaborate, and evaluate." It establishes a convergence innovation learning process that fosters active student engagement and deep understanding. Grounded in cognitive psychology and constructivist learning theory, this approach encourages students to actively construct knowledge through hands-on experience, experimental inquiry, and reflective discussion (see Figure 1). By introducing student-centered strategies, such as project-based learning, problem-based learning, and phenomenon-based learning, educators design activities tailored to students' interests, experience, and needs. This can stimulate enthusiasm for interdisciplinary inquiry while cultivating critical thinking, problem-solving, and innovative capacity (Dai, Zhang, et al., 2023). This approach lays a solid foundation for student development and contributes to innovative solutions for global challenges.

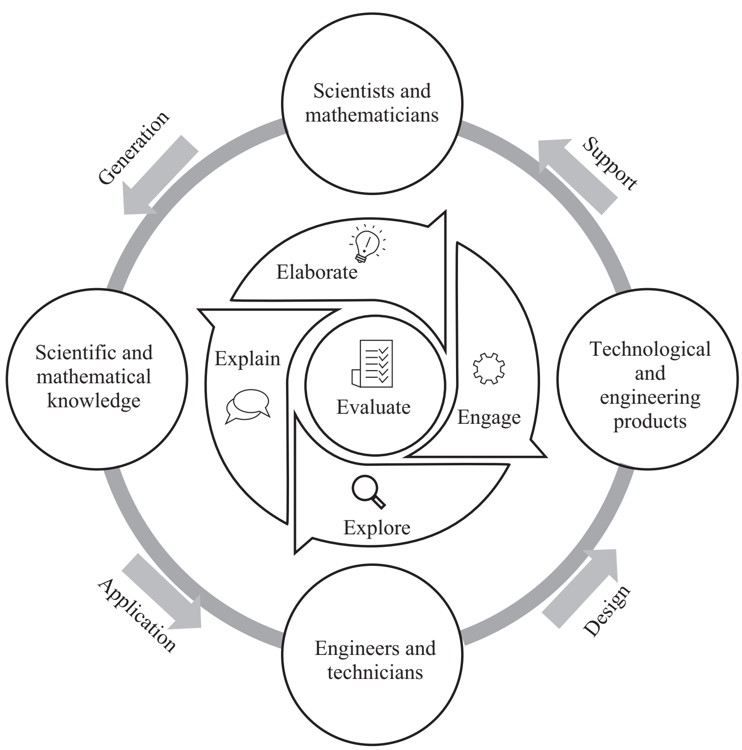

Figure 1 Logical Relationships among Various Elements in STEM Education

In addition, Smith and McCarty (2016) proposed a global transdisciplinary framework based on the convergence innovation theory. It synthesizes diverse perspectives from global research and advocates a more comprehensive, flexible, and inclusive methodology for exploring global problems. It seeks to transcend the limits of disciplinary and interdisciplinary approaches by examining issues from multiple angles and levels and promoting more integrated, interconnected scholarly practice. Beyond elements such as complex-problem identification, cross-domain methodological integration, and multi-party participation, the framework prioritizes decentering and de-sectoralization while integrating marginalized perspectives, fostering knowledge-sharing and co-creation, and applying cross-cultural, cross-epistemic, and cross-methodological approaches to explore solutions to broader global challenges.

3.2 Cultivating Creativity: The Central Axis of Convergence Innovation Education

Innovation, as the core of convergence innovation education, is a key driver of advances in new quality productive forces and the creation of new sources of economic growth. Fundamentally, innovation stems from the effective exercise of creativity: It is the application and realization of creative capacity. Creativity serves as the starting point of innovation and the motivating force throughout the innovation process. It denotes the ability to generate novel and valuable ideas within a given domain. Bono (1995) argued that creativity can be learned and practiced, and it involves breaking out of established patterns in order to look at things in a different way. Fleming and Marx (2006) described that creativity is the power to connect the seemingly unconnected. Creativity rarely emerges from a single mode of thought or within a single discipline.

Convergence innovation education, characterized by its interdisciplinary and integrative pedagogy, presents distinct advantages in contemporary educational practice, primarily through the cultivation of students' creative thinking. By combining diverse teaching methods with rich practical cases, it links abstract theory to concrete real-world contexts, thereby deepening students' understanding of knowledge and promoting its long-term retention. This approach motivates students to analyze problems from multiple perspectives and to explore innovative solutions, cultivating students' creativity, critical thinking, self-expression, and practical operational skills. For example, a course series on "paper waste" involves learning the history and production of paper while enabling students to understand and tackle environmental issues from several angles through statistical analyses in mathematics and poetry writing in language courses. This approach strengthens connections across domains, stimulating innovative thinking and enhancing hands-on skills and critical reasoning as students deal with real-world problems. By organizing instruction around a unifying theme, the faculty promotes interdisciplinary integration and creates a challenging, innovation-rich learning environment that nurtures students capable of in-depth analysis of complex issues and solving practical problems.

Some scholars have proposed five creative habits of mind: inquisitive, persistent, imaginative, disciplined, and collaborative, each linked to specific behavioral traits. Being inquisitive leads students to explore the unknown and challenge established assumptions; being persistent equips them with the resilience for addressing challenges and the courage for independent thinking; being imaginative fuels innovative, non-linear thinking; being disciplined ensures rigor in reasoning and the refinement of skills; and being collaborative unlocks the power of teamwork and collective intelligence (A New Direction, 2023). Collectively, these five habits cultivate a multifaceted, dynamic creative-thinking framework, fostering students' creative thinking to be broad, deep, and systematic.

3.3 The Pervasive Integration of AI in Convergence Innovation Education

The penetration of AI into education is demonstrating vast potential. Educational AI enables customization and adaptation of instructional content while fundamentally reforming educational assessment frameworks, student learning behavior analytics, and feedback mechanisms (Dai, Zhao, et al., 2023). First, AI-empowered precise data analysis and optimization of learning pathways provide educators with novel instructional strategy tools, making personalized learning a practicable reality. AI-empowered adaptive learning systems deliver tailored resources and feedback based on each learner's progress and comprehension, thereby achieving genuine individualized instruction. Second, AI applications in automated grading and feedback relieve educators of onerous assessment tasks. This enhances the objectivity and timeliness of assessment while enabling the faculty to attend to students' individual development and to design higher-value teaching activities. Moreover, AI enables the creation of engaging and immersive virtual learning environments that boost enjoyment and participation. More importantly, it facilitates collaborative learning and deep exploration of knowledge, establishing interactive platforms for knowledge sharing among learners across time and space. In student information management and risk early warning, AI further enhances decision support for education. By analyzing student behavior, academic performance, and related data, it promptly detects potential learning difficulties as well as moral and ethical risks, providing a scientific decision-making basis for educational administrators.

Large language models (LLMs) such as ChatGPT serve multiple roles in educational practice (Dai, Hu, et al., 2023). As a "possibility engine" (facilitating the discovery of a range of possibilities), it stimulates thought and supports creative expression. As a "Socratic debater," it challenges students' arguments and promotes deeper reflection. As a "collaborative coach," it aids group inquiry and problem solving. As a "guide on the side," it serves as a learning navigator to help students understand complex concepts. As a "personal tutor," it provides immediate and individualized feedback on learning progress. As a "co-designer," it participates in instructional design to transform content from unstructured to structured and ultimately to intelligent forms. As a "science museum," it offers rich experimental environments and interactive experiences. As a “learning partner,” it engages with students in knowledge reflection. As a “motivator,” it utilizes games and challenges to spark students’ interest in learning.

The Intelligent Education Algorithm Evaluation team at East China Normal University (ECNU) has developed a trustworthiness evaluation platform for intelligent education (EduTEP). The platform released the inaugural CALM-EDU evaluation framework and the CALM-EDU001 evaluation report for primary and secondary education from the dimensions of teaching knowledge, student development, and content knowledge. The report indicates that GPT-4 and other LLMs demonstrate outstanding overall performance on education-related applications and tasks (ECNU, 2023a). Subsequently, ECNU introduced EduChat, an education-domain chatbot system built on LLMs designed to support personalized, equitable, and empathetic intelligent education. EduChat exhibits four principal educational capabilities: open-ended question answering, automated essay scoring, heuristic instruction, and emotional support. Through pre-training based on a large corpus of education-specific texts and meticulous fine-tuning with high-quality customized instruction sets, EduChat delivers accurate knowledge responses and in-depth analyses, employs Socratic questioning to stimulate students’ critical thinking, and provides emotional and psychological support. The development of EduChat provides robust technical support for personalized and efficient intelligent education while offering new practical directions and research perspectives for deploying large models in the education sector (ECNU, 2023b).

The interdisciplinary integration of AI with other fields can be categorized into three levels (Xiao, 2022). The first level is integration with engineering and medicine, where AI functions as a highly efficient tool that strengthens applied practice. For example, the use of image recognition in disease diagnosis. The second level is integration with brain science. The development of AI has been inspired by neuroscience, with the rise of deep learning and neural networks serving as prime examples. The third level is integration with the humanities and social sciences. This is a bidirectional, comprehensive, and deep integration characterized by productive complementarity. AI enriches its comprehension of social phenomena by assimilating humanistic knowledge. Meanwhile, the humanities and social sciences employ AI to explore new epistemic frontiers, which promotes the joint advance of ideas and technology (Xiao, 2022). To accommodate the development of convergence innovation education under the penetration of AI, HEIs worldwide are actively experimenting with diverse educational models. Examples include Yonsei University (Republic of Korea), which established an AI education-centered, integrated intensive major; Petrozavodsk State University (Russia), which proposed a convergent educational model based on intelligent learning environments; Stanford University, which compiled the *Artificial Intelligence Teaching Guide*; and Ajou University (Republic of Korea), which established a research group on next-generation, super-intelligence network integrated education. These initiatives illustrate that deep integration of AI and education has emerged as a significant driver of academic innovation and educational reform.

4 Strategies and Recommendations

4.1 Integrating Convergence Innovation Research with Convergence Innovation Education to Cultivate Convergence Innovation Talent

The development of new quality productive forces urgently requires cultivating talent capable of integrating specialized knowledge and skills into specific fields, continuously contributing new ideas, theories, methods, and innovations, thereby generating outstanding economic, social, and ecological value. Singh (2016) observed that higher education is delivering a 20th-century education to 21st-century youth and preparing them for an unknown society. This mismatch between supply and demand in higher education calls for building more flexible, inclusive, and research-integrative higher education systems (Evans & Bahrami, 2020). It precisely identifies and supports students' individualized talents, rigorously fostering their creative and critical thinking, and leveraging technological means to strengthen students' decision-making capabilities and innovation competencies.

Reconceptualizing higher education for convergence innovation requires moving beyond piecemeal reforms toward systemic restructuring (Bainbridge & Roco, 2016a). Such restructuring operates along multiple dimensions simultaneously: (1) temporally, by extending educational engagement across the lifespan course rather than confining it to discrete institutional phases; (2) epistemically, by dissolving artificial boundaries between STEM fields and the humanities to enable integrative knowledge application; (3) relationally, by strengthening linkages between academic inquiry and socio-economic problem solving; and (4) methodologically, by blending formal instruction with informal, personalized learning pathways. These dimensions are not independent reform agendas. Instead, they are interdependent facets of a coherent transformation. Pursuing any single dimension in isolation risks generating fragmented initiatives that fail to achieve systemic impact. Therefore, the cultivation of convergence innovation talent demands institutional architectures capable of orchestrating simultaneous movement along all four dimensions.

Sharma and Sharma (2021) proposed an innovation framework for excellence in HEIs. This framework positions innovation as its driving force, aiming to achieve academic excellence through the interaction among various elements of HEIs while enhancing their contribution to society. It involves conceptual innovation, concept implementation, quality innovation, team roles, innovation in teaching and learning processes, administrative process innovation, cooperation process innovation, development process innovation, and value creation process innovation. That is, by fostering an open mindset and an interdisciplinary cooperative culture, it increases the rate of creative transformation, strengthens teamwork, updates teaching methods, optimizes administrative management, and promotes inter-institutional cooperation, thereby fully realizing the value of innovation as the core asset of HEIs. Technological iteration is indispensable to this process. To reveal the complexity of digital intelligence-powered educational innovation, Leurs (2018) constructed a multidimensional innovation ecosystem based on four innovation spaces, namely, intelligence space, solution space, technology space, and talent space. The intelligence space is devoted to collecting and analyzing information from both inside and outside the education sector, offering objective interpretations and theoretical constructs of the current situation to provide a scientific foundation for decision-making. The solution space focuses on empirical research and the incubation of solutions, promoting prototype testing, rapid trial-and-error, and iterative model refinement to optimize educational models and practices. The technology space reflects the driving role of information technology and big data in educational innovation, aiming to advance the transformation of learning tools and platforms. The talent space emphasizes systematic talent development and organizational capacity building to enable change, encompassing leadership development, team building, and the cultivation of an innovation culture. These four spaces are mutually supportive yet functionally distinct. They guide HEIs, in an era of complex change, to leverage innovative tools and systems thinking to continuously advance academic excellence and institutional innovation.

Against this backdrop, convergence innovation research and convergence innovation education substantially strengthen students' compound abilities as follows (Pearce et al., 2018): (1) systems thinking, which enables students to understand complex problems holistically and to address them through comprehensive approaches; (2) innovation, which encourages students to transcend traditional frameworks and to explore and implement novel ideas and solutions; (3) communication and collaboration, which strengthens students' ability to communicate and engage in teamwork effectively across diverse cultural and disciplinary contexts; (4) critical thinking, which cultivates students' capabilities for independent analysis and evaluation and the habit of making reasoned judgments; and (5) lifelong learning, which equips students to continue developing and updating themselves in rapidly changing environments.

In higher education, it is imperative to establish sustainable mechanisms and a diverse, coordinated training system to effectively nurture students' innovative capabilities. For example, some HEIs established dedicated units for innovation, creation, and entrepreneurship, including educational centers, practice centers, and research centers (Qian & Zhang, 2020). Moreover, they joined forces with industry enterprises, peer HEIs, society, and educational development foundations to pool superior resources. Through the mechanism of extensive consultation, joint contribution, and shared benefits, build an innovation–creativity–entrepreneurship education system aimed at producing innovative talent who meets enterprise recruitment needs and broader societal development goals. In addition, they promote in-depth exchanges and cooperation between the academic community and the industry through convergence innovation research and interdisciplinary research methods. They also explore forward-looking and applicable solutions to interdisciplinary issues to enhance students' systems thinking and innovative practical abilities for the purpose of addressing complex problems. Furthermore, they establish an integrated and hierarchical modular teaching model encompassing "teaching, learning, competition, and creation" based on the hierarchical and progressive course design approach. Four teaching stages are proposed to cultivate innovative talent, namely, learning through teaching, practice during learning, creating in practice, and transferring in creation. They secure premium resources from industry, enterprises, and society, and establish three-dimensional cooperation platforms by integrating universities and enterprises, coordinating among universities, and conducting Sino-foreign educational cooperation. Therefore, a collaborative educational community can be formed, characterized by complementary advantages, joint project development, shared achievements, and mutual benefit. Moreover, they strengthen students' practical application skills as well as innovation and entrepreneurship capabilities, and cultivate high-caliber, application-oriented, innovative talent who adapt to regional economic and industrial development. An innovative talent training system can be jointly built from aspects such as course systems, teaching models, practical platforms, and service systems.

4.2 Fostering a Convergence Innovation Culture Grounded in SciTS

The science of team science (SciTS) is an emerging field that examines the organizational structures, communication patterns, and behavioral processes of scientific research teams, with the aim of identifying the key factors that facilitate or impede interdisciplinary research (Stokols, Hall, et al., 2008). This domain highlights that experts with various disciplinary backgrounds in a research team leverage their collective wisdom to address sophisticated scientific and societal challenges that cannot be tackled by a single discipline (Stokols, Misra, et al., 2008). By fostering in-depth collaboration among individuals, SciTS catalyzes a paradigmatic shift in scientific inquiry, that is, from the traditional model of isolated individual investigators toward a more open, collaborative, and interconnected approach. It facilitates the resolution of complex problems in contemporary scientific research and provides novel perspectives and methods for scientific knowledge creation, dissemination, and application, thereby serving as a central driver of scientific progress and innovation.

The convergence innovation culture, embedded in organizations and teams, emphasizes the blending of diverse knowledge, skills, and perspectives to address problems collaboratively. In such an environment, individuals from heterogeneous backgrounds freely exchange ideas, stimulate innovation, spark insights, and jointly generate new knowledge, technologies, and solutions. The process of converting research findings into new quality productive forces encompasses need identification, analysis, dissemination, stakeholder engagement, R&D, experimentation, and eventual implementation. Sole reliance on individual effort impedes organized research and the effective technology transfer. Building a convergence innovation culture grounded in SciTS entails transforming scattered, temporary, short-term cooperation into organized, sustained collaborations. This facilitates the transition of mechanistic organizations into emergent forms, thereby counteracting organizational rising entropy.

To cultivate a convergence innovation culture grounded in SciTS, it is imperative for HEIs to pursue coordinated measures across multiple dimensions such as strategic planning, organizational design, talent development, and evaluation and incentives. Firstly, break down disciplinary barriers and create platforms for multidisciplinary exchanges and cooperation through institutional design and policy support. This encompasses establishing interdisciplinary research centers, offering interdisciplinary courses and programs, allocating dedicated interdisciplinary research funds, and setting evaluation criteria for collaborative research projects as well as recognition and reward mechanisms for interdisciplinary team results. Secondly, it is essential to cultivate team members' interdisciplinary thinking and collaborative capabilities and foster openness and a cooperative ethos. Teachers are advised to facilitate students' active engagement in interdisciplinary team projects and practical activities. Through practice-based learning and problem solving, they can appreciate the significance of collaborative work, recognize the value of diverse disciplinary perspectives, and therefore cultivate convergence innovation-oriented talent. Thirdly, promote lateral leadership and distributed governance to enhance organizational flexibility and innovative capabilities, encourage information sharing, devolve authority, and enable autonomous decision-making, thereby creating a supportive environment for interdisciplinary collaboration. Fourthly, implement dynamic team formation and cross-boundary mobility. To elaborate, adopt project requirement-based, flexible team composition strategies that adjust membership according to research directions and project objectives, while encouraging the mobility of scholars and students across disciplines and research teams to facilitate the exchange of knowledge and experience. Fifthly, optimize performance evaluation and incentive systems by designing incentive mechanisms aligned with a convergence innovation culture and establishing a fair, comprehensive evaluation framework that fosters interdisciplinary collaboration. It is crucial that the framework transcends traditional single-discipline-outcome-centric models and incorporates indicators, such as teamwork, interdisciplinary research engagement, and societal impact, thereby ensuring appropriate recognition and reward for the achievements of interdisciplinary research teams.

Moreover, by cultivating cultural values that privilege interdisciplinary collaboration, fostering an open and sharing organizational climate, reinforcing team spirit and collective identity, and promoting tolerance for failure and experimentation, HEIs are well-placed to enhance the quality and effectiveness of academic innovation and societal engagement, providing robust intellectual support and solutions to global challenges.

4.3 Enhancing the Governance System for Convergence Innovation in Higher Education

Governance refers to the collective capacity for achieving socially desired community benefits in complex and changing circumstances, highlighting the interaction of diverse mechanisms. It represents a multidimensional, multi-perspective process of convergent action. Convergence governance, as an interdisciplinary, cross-boundary governance approach, seeks to construct efficient, equitable, and sustainable development models by integrating diverse perspectives and resources from science, technology, and social governance. Different from top-down governing, the convergence governance process is dominated by horizontal links and self-organization principles. It has the potential to catalyze fundamental transformations across scientific, technological, and social domains. The U.S. nanotechnology governance strategy aims to achieve transformative, responsible, and inclusive development (Roco, 2008). The governance model associated with Elon Musk and his company, SpaceX, converges diverse societal stakeholders and continually advances the development of novel convergent technologies independent of governmental functions (Roco, 2020). The emergence of participatory governance instruments, such as games, co-designs, and social media, expands avenues for public engagement in the governance of science and education, enabling technological and educational trajectories to be more effectively aligned with societal needs and values. Regarding the regulation of convergent technologies, two core regulatory approaches should be prioritized (Roco & Bainbridge, 2013). Firstly, an exploration of the scalability of existing regulatory frameworks should be conducted. Secondly, innovative, exploratory, and flexible regulatory and governance models need to be developed.

Guided by this, the Korea Institute of Science and Technology (KIST) established a center for convergence research policy. It applies convergence concepts to coordinate decision-making across government levels and holistically accounts for critical factors in research and development projects and investment decisions in order to promote long-term national development and strengthen capacities for exploring future technologies. Moreover, early spaceflight platforms, Silicon Valley, and semiconductor research consortia have demonstrated the practical effectiveness of convergence governance across diverse domains.

Convergence governance in higher education is critical for enabling cross-boundary collaboration that drives educational innovation and development. The World Economic Forum (2022) identified four major trends shaping the future of higher education:

First, ubiquitous learning. It signifies the convergence of virtual and physical classrooms, on-campus and off-campus learning, immersive and experiential modalities, and the transmission of knowledge with its practical application. Second, active learning. Effective learning relies on principles such as spaced learning, emotional learning, and the application of knowledge. Third, skills learning. It is necessary for teachers and students in higher education to continually update their teaching and learning skills to respond to evolving future demands, and acquiring these skills typically necessitates interdisciplinary competencies. Fourth, formative assessment. It entails integrating formal and informal assessments, self-assessment, and peer-assessment, as well as evaluations of competencies and project work. It emphasizes the learning process itself rather than solely focusing on final outcomes, aiming to foster students' continuous improvement and development.

It is the responsibility of higher education to recognize and address the unanticipated challenges arising in transformative learning processes and to leverage mechanisms inherent in complex dynamic systems to advance convergence innovation governance. Governance frameworks for convergence innovation in higher education coalesce around several interrelated imperatives (Vanasupa et al., 2014). At the structural level, dismantling disciplinary silos and instituting mechanisms for intellectual cross-fertilization constitute necessary preconditions. At the pedagogical level, adaptive curriculum architectures need to replace rigid programmatic structures to equip students for uncertain futures. At the environmental level, research cultures that tolerate experimental failure and protect scholarly autonomy prove essential for sustaining innovative inquiry. These imperatives converge on a common requirement: aligning theoretical instruction with practical application through robust HEI-industry partnerships while ensuring continuous faculty development. The governance challenge lies not in implementing any single pathway but in orchestrating their simultaneous pursuit within coherent institutional frameworks. Petersen et al. (2021) advocated advancing interdisciplinary and cross-sectoral research by foregrounding a shared purpose (Why do I participate?), practice (How do we interact?), and place (Where do we interact?), and catalyzing innovation and collaboration within HEIs by establishing "undisciplinary centers." It is evident that the establishment of open, collaborative platforms and physical spaces is indispensable to guiding the sustainable advancement of convergence innovation research. They aim to cultivate a supportive work environment and provide a practical approach, rather than to function merely as formal entities. Furthermore, Dubé (2014) proposed a polycentric governance model as a strategy for addressing the challenges posed by complex systems, emphasizing interactions and collaborations among individuals, communities, markets, and the state. Through a multi-level governance architecture, the model seeks to strengthen the self-organizing and adaptive capacities of governing entities, with the aim of fostering a dynamic, flexible, and inclusive governance environment.

In summary, constructing a forward-looking, inclusive, and holistic system of convergence innovation education is indispensable for higher education to adapt to epochal change and to take a leading role in social innovation and development. Convergence innovation education extends beyond interdisciplinary engagement, highlighting horizontal and vertical integration across distinct domains, such as technology and the arts, as well as engineering and social sciences, and across multiple levels. It prioritizes the synthesis of theory and practice, humanistic concern and technological advancement, global perspective and local praxis, and individual development and cross-sector collaboration. Its central value lies in cultivating talent endowed with adaptability, creativity, and global competitiveness. Against the pervasive impact of technologies such as AI, the mission of higher education, which is to pursue excellence and to drive new quality productive forces, has become more critical and urgent. It is imperative for HEIs to fully appreciate the holistic, systemic, design-oriented, and emergent character of convergence innovation and, with an open and inclusive perspective, to fulfill its pivotal role in the national effort to build a leading country in education.

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